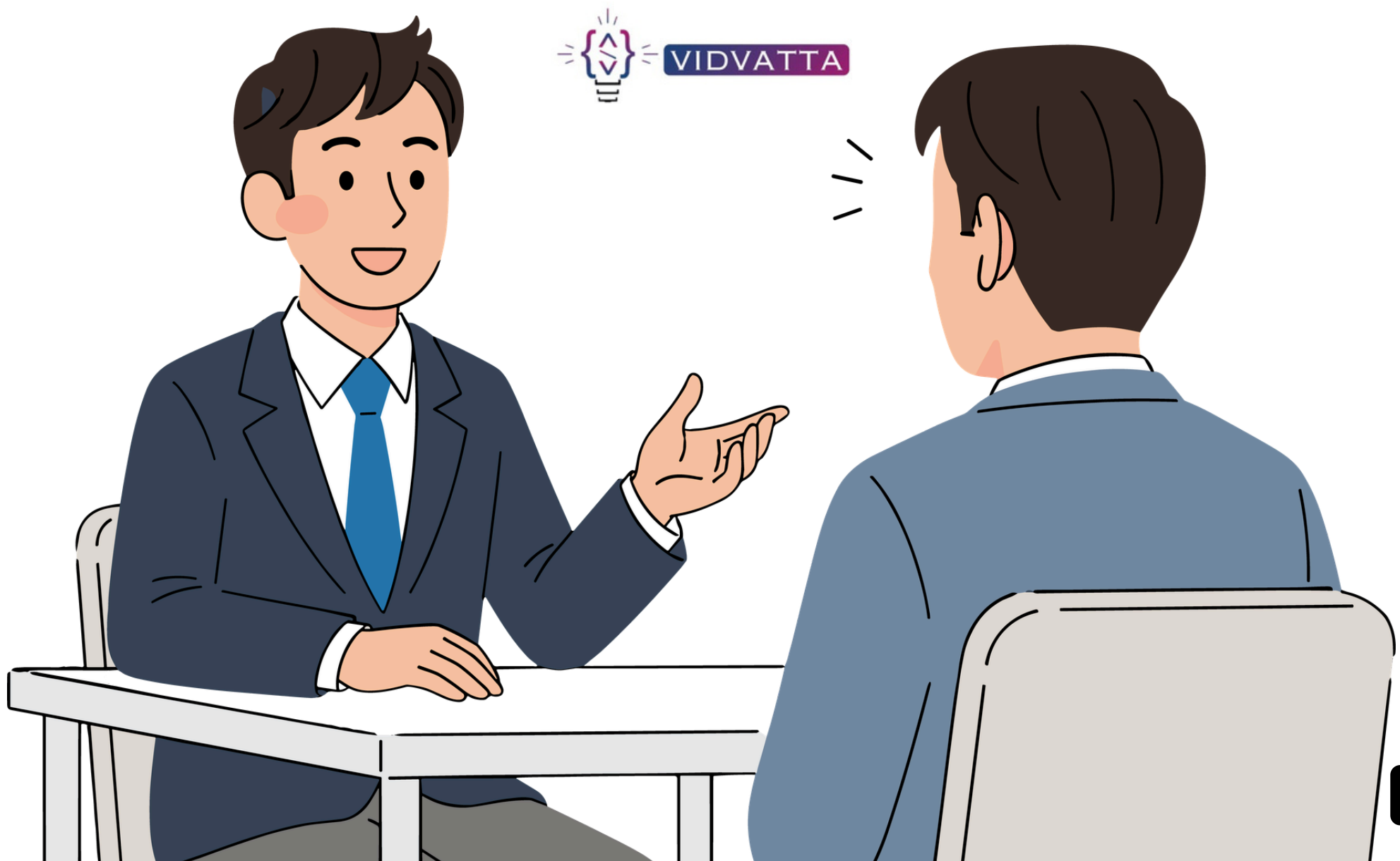


AI ENGINEERING INTERVIEW QUESTIONS

BEGINNER, INTERMEDIATE, ADVANCED



Next



Want to switch to AI/ML but don't know where to start? DM "Guide" for expert direction.



BEGINNER LEVEL

1. What is Artificial Intelligence?

- AI gives machines the ability to perform tasks that require human intelligence.
- Involves learning, reasoning, understanding language, speech, and visual perception.
- Used in applications like recommendation systems, chatbots, and image classification.

2. What is supervised vs. unsupervised learning?

- Supervised uses labeled data, mapping inputs to known outputs during model training.
- Unsupervised uses unlabeled data, finding structures or patterns without explicit answers.
- Supervised: classification/regression tasks; Unsupervised: clustering, dimensionality reduction applications.

Next



Trying to move into AI but stuck?



3. What is overfitting and how can you prevent it?

- Overfitting happens when a model learns noise, failing on new data.
- Use L1/L2 regularization or restrict model complexity to control overfitting.
- Employ cross-validation or dropout for robust and generalizable models.

4. Name some key Python libraries used in AI/ML.

- NumPy and Pandas handle numerical computation and data manipulation efficiently.
- Scikit-learn offers machine learning algorithms for quick experimentation and evaluation.
- TensorFlow and PyTorch support building and training deep neural networks.

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5. What is a neural network?

- It's a mathematical model mimicking the brain's interconnected neuron structure.
- Consists of input, hidden, and output layers to transform and learn from data.
- Used extensively for image, text, and speech data modeling tasks.

6. What is data normalization, and why is it important?

- Normalization scales input features to a fixed range, typically 0 to 1.
- Prevents features with larger scales from dominating learning during model training.
- Essential for improving convergence speed and stability in neural network optimization.

Next





INTERMEDIATE LEVEL

1. What is the bias–variance trade-off?

- High bias means underfitting, causing poor accuracy on both training and test data.
- High variance causes overfitting, sensitivity to training data noise, and poor generalization.
- The challenge is finding optimal complexity for good performance on unseen examples.

2. Difference between batch gradient descent & stochastic gradient descent ?

- Batch gradients use the entire data set for each weight update, requiring more memory.
- SGD updates weights after every single sample, introducing noise for faster convergence.
- Mini-batch balances both approaches, updating weights using small subsets per iteration.

Next



3. What is dropout and why is it used?

- Dropout randomly disables neurons during training, reducing over-reliance on specific features.
- Encourages the network to learn redundant representations and robust features.
- Helps to improve test accuracy and prevent neural network overfitting.

4. How do you handle missing data in a dataset?

- Remove rows or columns with excessive missing values if they compromise data quality.
- Impute missing values using feature mean, median, mode, or predictive models.
- Flag missingness as a feature for certain machine learning algorithms to leverage.

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View Details



5. What is transfer learning?

- Uses knowledge from one task's model to boost performance on a related task.
- Fine-tunes a pre-trained model on new data, saving time and computational resources.
- Commonly used in deep learning for image recognition and natural language processing.

6. What is feature scaling, and why is it important?

- Feature scaling standardizes input variables to a uniform scale or distribution.
- Improves training stability and convergence speed, especially for gradient descent-based algorithms.
- Prevents dominance of features with larger scales during model learning and optimization.

Next



ADVANCED LEVEL

1. Explain vanishing and exploding gradients in deep networks.

- Vanishing gradients halt learning in early layers due to excessively small updates.
- Exploding gradients cause numerical instability from overly large weight updates.
- Solve with ReLU activations, residual connections, proper initialization, or gradient clipping.

2. How does attention work in transformers?

- Attention calculates weights denoting importance of each word to others in a sequence.
- Uses Key, Query, and Value vectors to dynamically focus on relevant input parts.
- Enables transformers to understand context and long-range dependencies better.



3. What are differences between LSTM and GRU?

- LSTM uses three gates for more complex memory control, handling sequences and dependencies.
- GRU employs only two gates, resulting in fewer parameters and faster computations.
- Performance is often similar, but GRUs are simpler and more resource-efficient.

4. How would you deploy a trained ML model in production?

- Serialize the model with joblib, pickle, TorchScript, TensorFlow SavedModel, or ONNX format.
- Serve predictions via Flask, FastAPI, or cloud APIs within a Docker container.
- Deploy on cloud platforms (AWS Sagemaker, GCP AI Platform) for scalable inference.

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5. What are GANs and how do they work?

- GANs use two networks, generator and discriminator, in a competitive learning process.
- Generator creates data; discriminator distinguishes between real and fake inputs.
- Both improve as generator fools discriminator, achieving realistic data generation outputs.

6. How do you handle imbalanced datasets in classification?

- Use resampling techniques: oversampling minority or undersampling majority class for balance.
- Apply penalized algorithms or class weighting in the loss function during training.
- Employ evaluation metrics like F1-score, precision-recall curve over simply accuracy.

Next



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